The Dyson Equation for Random Matrices

Olivia Choi, Elliott Kobelansky (Mentor: Kevin Xiao)

The Dyson equation, developed in recent years, provides a unified framework for studying the spectral properties of large random matrices. We introduce core tools of random matrix theory: the empirical spectral distribution, the Stieltjes transform, and the resolvent, and use them to derive the simplest instance of the Dyson equation for Wigner matrices, recovering the Wigner semicircle law.

1 Introduction

Random matrix theory was first developed in the 1950s by Eugene Wigner to model the energy levels of heavy atomic nuclei. Since then, it has grown into a central area of probability and mathematical physics, with deep connections to combinatorics, number theory, and even the distribution of zeros of the Riemann zeta function. More recently, it has found applications in statistics, optimization, and machine learning.

A major focus of the field is the asymptotic behavior of eigenvalues of large random matrices. In classical probability, foundational results such as the Law of Large Numbers and the Central Limit Theorem describe universal behavior for sums of independent random variables. However, these results do not directly extend to the non-commutative setting of matrices, where the primary objects of interest are spectra rather than scalar averages. This raises a central question in random matrix theory: to what extent do analogous universality principles hold for eigenvalue distributions?

Much of the theory is developed in the setting of self-adjoint matrices. This assumption is motivated both by applications, particularly in physics, where observables are represented by Hermitian operators, and by mathematical convenience. Self-adjoint matrices have real eigenvalues, which allows spectral analysis to remain within the real line and significantly simplifies the probabilistic framework.

While the general theory presented extends to complex Hermitian matrices with only minor modifications, we restrict attention to the real symmetric case for clarity.

Definition 1.1 (Symmetric Random Matrix). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. An $N \times N$ symmetric random matrix is a measurable mapping

$$H : \Omega \to \mathrm{Sym}_N(\mathbb{R}) := \{A \in \mathbb{R}^{N\times N} : A^T = A\}.$$

Equivalently, H may be specified by a collection of real-valued random variables $H_{ij} : \Omega \to \mathbb{R}$ for $1 \le i, j \le N$, such that each H_{ij} is measurable and $H_{ij} = H_{ji}$ almost surely for all $1 \le i, j \le N$.

2 Spectral Methods

In this section, we introduce the foundational mathematical tools to analyze the spectral properties of random matrices. These tools are necessary to link algebraic matrix properties to probability measures.

2.1 Empirical Spectral Distribution

Definition 2.1 (Empirical Spectral Distribution). Let $H \in \mathbb{R}^{N\times N}$ be a symmetric matrix with eigenvalues $\lambda_1, \ldots, \lambda_N$. The *empirical spectral distribution (ESD)* of H is the probability measure

$$\mu_N = \frac{1}{N}\sum_{i=1}^{N} \delta_{\lambda_i},$$

where $\delta_x(A) = \mathbf{1}_A(x)$ is the Dirac measure at x. If H is random, then μ_N is a random probability measure.

Of particular interest is the asymptotic behaviour of the sequence of random measures $(\mu_N)_{N\geq 1}$ associated with a given sequence of random matrices $(H_N)_{N\geq 1}$, in particular whether μ_N converges weakly (in an appropriate probabilistic sense) to a deterministic limiting measure as $N \to \infty$.

2.2 The Stieltjes Transform

The Stieltjes transform provides an analytic encoding of a probability measure, allowing weak convergence of measures to be studied via convergence of analytic functions. It is the analogue in random matrix theory of the characteristic function in probability theory, in that it encodes a probability measure in a transform domain where convergence is more tractable.

Definition 2.2 (Stieltjes Transform). Let μ be a Borel probability measure on $\mathbb{R}$. The *Stieltjes transform* of μ is the function

$$m_\mu(z) = \int_{\mathbb{R}} \frac{d\mu(x)}{x - z},$$

defined for $z \in \mathbb{H}$, where $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$. If μ is random, then m_μ is a random complex-valued function.

It can be shown that the Stieltjes transform is analytic, and has $\operatorname{Im} m_\mu(z) > 0$ for all $z \in \mathbb{H}$.

Proposition 2.3 (Inverse Stieltjes Transform). *Let μ be a Borel probability measure, $a, b \in \mathbb{R}$ with $a < b$. Then,*

$$\lim_{\eta\to 0^+} \frac{1}{\pi}\int_a^b \operatorname{Im} m_\mu(x + i\eta)\, dx = \mu((a,b)) + \frac{1}{2}\big(\mu(\{a\}) + \mu(\{b\})\big).$$

Proof. First, observe that

$$\operatorname{Im} m_\mu(x+i\eta) = \int_{\mathbb{R}} \operatorname{Im}\left(\frac{1}{t-(x+i\eta)}\right) d\mu(t) = \int_{\mathbb{R}} \frac{\eta}{(t-x)^2+\eta^2}\, d\mu(t).$$

Since the integrand is non-negative, Tonelli's theorem gives

$$\int_a^b \operatorname{Im} m_\mu(x+i\eta)\, dx = \int_{\mathbb{R}} \int_a^b \frac{\eta}{(t-x)^2+\eta^2}\, dx\, d\mu(t).$$

Substituting $u = (t-x)/\eta$ into the inner integral yields

$$\int_a^b \frac{\eta}{(t-x)^2+\eta^2}\, dx = \int_{(a-t)/\eta}^{(b-t)/\eta} \frac{1}{1+u^2}\, du = \arctan\left(\frac{b-t}{\eta}\right) - \arctan\left(\frac{a-t}{\eta}\right).$$

Thus, we obtain

$$\int_a^b \operatorname{Im} m_\mu(x+i\eta)\, dx = \int_{\mathbb{R}} \left[\arctan\left(\frac{b-t}{\eta}\right) - \arctan\left(\frac{a-t}{\eta}\right)\right] d\mu(t).$$

The integrand is bounded by π, hence integrable with respect to μ. Moreover, as $\eta \to 0^+$,

$$\arctan\left(\frac{b-t}{\eta}\right) - \arctan\left(\frac{a-t}{\eta}\right) \to \pi \mathbf{1}_{(a,b)}(t) + \frac{\pi}{2}\big(\mathbf{1}_{\{a\}}(t) + \mathbf{1}_{\{b\}}(t)\big).$$

Finally, applying the dominated convergence theorem, we obtain the desired limit

$$\lim_{\eta\to 0^+} \frac{1}{\pi} \int_{\mathbb{R}} \left[\arctan\left(\frac{b-t}{\eta}\right) - \arctan\left(\frac{a-t}{\eta}\right)\right] d\mu(t) = \mu((a,b)) + \frac{1}{2}\big(\mu(\{a\}) + \mu(\{b\})\big).$$

□

In particular, observe that if μ is absolutely continuous with respect to the Lebesgue measure λ on $\mathbb{R}$, the inverse Stieltjes transform of m_μ recovers exactly $\mu(a,b)$. Furthermore, the density $\rho = \frac{d\mu}{d\lambda}$ has the property

$$\rho(x) = \lim_{\eta\to 0^+} \frac{1}{\pi} \operatorname{Im}\, m_\mu(x+i\eta),$$

almost everywhere, since $\operatorname{Im}\, m_\mu(x+i\eta) = (P_\eta * \rho)(x)$, where P_η is the Poisson kernel. Since $(P_\eta)_{\eta>0}$ is an approximation to the identity, a result in [SS05] implies that $(P_\eta * \rho)(x) \to \rho(x)$ for almost every $x \in \mathbb{R}$.

Proposition 2.4 (Equivalence of Convergence). *Let $(\mu_N)_{N\geq 1}$ be a sequence of Borel probability measures on $\mathbb{R}$, and let m_{μ_N} denote their Stieltjes transforms. Then $\mu_N \Rightarrow \mu$ for some probability measure μ if and only if*

$$m_{\mu_N}(z) \to m_\mu(z) \quad \textit{for all } z \in \mathbb{H}.$$

The proof of this result is found in [AGZ10] and requires several technical lemmas that are not pertinent to the scope of this report. Nonetheless, the result is fundamental as it reduces questions of convergence of measures to convergence of analytic functions.

In particular, when (μ_N) is a sequence of random probability measures, convergence in probability of the respective Stieltjes transforms m_{μ_N} to the transform m_μ of a limit measure m in probability for all $z \in \mathbb{H}$ implies that μ_N converges weakly to μ in probability.

2.3 The Resolvent

Finally, we introduce the resolvent matrix $G(z)$, which provides a convenient analytic representation of spectral information. Rather than working directly with the empirical eigenvalue distribution, which is difficult to access through the characteristic polynomial, we instead study the resolvent whose analytic structure allows access to the empirical spectral distribution through algebraic manipulation.

Definition 2.5 (Resolvent). Let $H \in \mathbb{R}^{N\times N}$ be a symmetric matrix. The resolvent of H is the matrix-valued function

$$G(z) = (H - zI)^{-1},$$

defined for all $z \in \mathbb{C}$ such that z is not an eigenvalue of H.

Proposition 2.6 (Resolvent-Stieltjes Relation). *Let $H \in \mathbb{R}^{N\times N}$ be a symmetric matrix. Then, for all $z \in \mathbb{H}$ for which $G(z)$ is defined,*

$$\frac{1}{N}\operatorname{tr} G(z) = m_N(z),$$

where m_N is the Stieltjes transform of the empirical spectral distribution of H.

Proof. Since H is real symmetric, it admits an orthogonal diagonalization $H = Q\Lambda Q^T$, where $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_N)$ and $QQ^T = I$. For all $z \in C$ for which $G(z)$ is defined, we have

$$G(z) = (H - zI)^{-1} = (Q\Lambda Q^T - zQIQ^T)^{-1} = Q(\Lambda - zI)^{-1}Q^T.$$

Using the cyclic invariance of the trace,

$$\frac{1}{N}\operatorname{tr} G(z) = \frac{1}{N}\operatorname{tr}[(\Lambda - zI)^{-1}Q^TQ] = \frac{1}{N}\operatorname{tr}(\Lambda - zI)^{-1} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\lambda_i - z}.$$

By definition of the empirical spectral distribution $\mu_N = \frac{1}{N}\sum_{i=1}^{N}\delta_{\lambda_i}$, its Stieltjes transform is

$$m_N(z) = \int_{\mathbb{R}}\frac{1}{x-z}\,d\mu_N(x) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\lambda_i - z} = \frac{1}{N}\operatorname{tr} G.$$

□

This connection allows spectral problems to be framed as analytical matrix problems involving $G(z)$, which is especially useful for determining the limiting spectral distribution of a sequence of matrices $(H_N)_{N\geq 1}$. By Proposition 2.4, it suffices to study the convergence of the normalized trace of the resolvent. If this limiting quantity exists and is deterministic, it can be inverted using Proposition 2.3 to recover the limiting spectral distribution.

At this point, we have reframed the problem of determining the limiting spectral distribution of $(H_N)_{N\geq 1}$ in terms of the normalized trace of the resolvent $G(z)$. Depending on the class of matrices (H_N), a variety of methods exist to establish convergence of the normalized trace of the resolvent. In many classical random matrix ensembles, the limiting Stieltjes transform is characterized as the solution of a deterministic self-consistent equation, commonly referred to as the Dyson equation. We turn to this formulation next.

3 The Dyson Equation for Wigner Matrices

Wigner matrices are a fundamental starting point in random matrix theory and arise as a simplified description of large complex systems with many weakly interacting components. Their importance stems from the fact that, despite the simplicity of the independence assumptions on the entries, they capture universal features of large random systems.

Definition 3.1 (Wigner Matrix). A *Wigner matrix* H is a symmetric random matrix $(H_{ij})_{i,j=1}^N$ where the entries H_{ij} are independent and identically distributed with

$$\mathbb{E}[H_{ij}] = 0, \quad \mathbb{E}[H_{ij}^2] = \frac{1}{N}$$

for $i \leq j$, and $H_{ji} = H_{ij}$ for $i > j$.

No further assumptions are imposed on the distribution of H_{ij}, it may be discrete or continuous, and need not satisfy any smoothness or structural conditions beyond the condition on the first and second moments. For examples of various Wigner matrices as well as simulations of their asymptotic distribution, see Section 4.1. These experiments suggest that the asymptotic distribution of (H_N) does not depend on the specific distribution of the entries, and is given by the semicircle distribution. The following theorem gives a precise statement of this convergence, via the Dyson equation for Wigner matrices. We expand the argument presented in [Erd19] by filling in several intermediate steps.

Theorem 3.2 (Wigner Semicircle Law via the Dyson Equation). *Let $(H_N)_{N\geq 1}$ be a sequence of Wigner matrices. There exists a deterministic function $m : \mathbb{H} \to \mathbb{H}$ such that $m_N(z) \xrightarrow{p} m(z)$ as $N \to \infty$ pointwise, where m_N is the Stieltjes transform of the empirical spectral distribution μ_N of H_N. Moreover, m is the unique solution to the Dyson equation*

$$m(z) = -\frac{1}{z + m(z)}.$$

Let μ be the probability measure whose Stieltjes transform is m. Then $\mu_N \Rightarrow \mu$ in probability, and μ is absolutely continuous with respect to the Lebesgue measure with density

$$\rho(x) = \frac{1}{2\pi}\sqrt{(4-x^2)_+}, \qquad x \in \mathbb{R}.$$

3.1 Probabilistic Step

The first objects of interest are the diagonal elements G_{ii} of the resolvent. For a fixed i, we may assume without loss of generality that $i = 1$, since the entries are i.i.d (up to symmetry), and relabeling the indices does not affect any relevant quantities. For convenience, we will often write $G(z)$ simply as G.

Lemma 3.3 (Schur Formula). *Let $A \in \mathbb{C}^{n\times n}$, $B \in \mathbb{C}^{m\times n}$, and $C \in \mathbb{C}^{m\times m}$, and define*

$$D = \begin{pmatrix} A & B^T \\ B & C \end{pmatrix}.$$

Assume C is invertible and define the Schur complement $D' = A - B^T C^{-1} B$. If D is invertible, then D' is invertible, and $(D^{-1})_{ij} = (D')^{-1}_{ij}$ for $i, j \le n$.

Proof. We factor the block matrix D using block Gaussian elimination,

$$\begin{pmatrix} I_n & -B^T C^{-1} \\ 0 & I_m \end{pmatrix} \begin{pmatrix} A & B^T \\ B & C \end{pmatrix} = \begin{pmatrix} A - B^T C^{-1} B & 0 \\ B & C \end{pmatrix} = \begin{pmatrix} D' & 0 \\ B & C \end{pmatrix}.$$

Factor once more on the right,

$$\begin{pmatrix} D' & 0 \\ B & C \end{pmatrix} \begin{pmatrix} I_n & 0 \\ -C^{-1}B & I_m \end{pmatrix} = \begin{pmatrix} D' & 0 \\ 0 & C \end{pmatrix}.$$

Combining the two factorizations yields

$$D = \begin{pmatrix} I_n & -B^T C^{-1} \\ 0 & I_m \end{pmatrix}^{-1} \begin{pmatrix} D' & 0 \\ 0 & C \end{pmatrix} \begin{pmatrix} I_n & 0 \\ -C^{-1}B & I_m \end{pmatrix}^{-1}.$$

Since the outer factors are invertible whenever C is invertible, we see that D is invertible if and only if D' is invertible. Furthermore,

$$D^{-1} = \begin{pmatrix} I_n & 0 \\ -C^{-1}B & I_m \end{pmatrix} \begin{pmatrix} D'^{-1} & 0 \\ 0 & C^{-1} \end{pmatrix} \begin{pmatrix} I_n & -B^T C^{-1} \\ 0 & I_m \end{pmatrix},$$

which has upper left block D'^{-1}. □

Let $H^{[i]}$ denote the (i,i) minor of H. Set $G^{[i]} = (H^{[i]} - zI)^{-1}$. Note that this is not the same as $((H-zI)^{-1})^{[i]}$. For the case of $i = 1$, we have

$$H - zI = \begin{pmatrix} H_{11} - z & a^T \\ a & H^{[1]} - zI_{N-1} \end{pmatrix},$$

where $a = (H_{12}, \ldots, H_{1N})^T$. Applying Lemma 3.3 gives

$$G_{11} = \left[(H - zI)^{-1}\right]_{11} = \frac{1}{H_{11} - z - a^T G^{[1]} a}.$$

Since a and $G^{[1]}$ are independent and $\mathbb{E}[H_{1j}H_{1k}] = \delta_{jk}$, we have that

$$\mathbb{E}[a^T G^{[1]} a \,\big|\, G^{[1]}] = \sum_{j,k>1} \mathbb{E}[H_{1j}H_{1k}] G^{[1]}_{jk} = \frac{1}{N-1} \operatorname{tr} G^{[1]}.$$

It is also possible to show that $\operatorname{Var}(a^T G^{[1]} a) = O(1/N)$, although its proof is technical and not essential to the main structure of the derivation and has therefore been omitted. Along with the fact that $\operatorname{Var}(H_{11}) = 1/N$, we have that

$$G_{11} = -\frac{1}{z + \frac{1}{N-1}\operatorname{tr} G^{[i]} + o_p(1)}.$$

Lemma 3.4. *Let $H \in \mathbb{R}^{N\times N}$ be symmetric with eigenvalues $\lambda_1 \geq \ldots \geq \lambda_N$. Let $\gamma_1 \geq \ldots \geq \gamma_{N-1}$ be the eigenvalues of the (i,i) minor $H^{[i]}$. Then,*

$$\lambda_1 \geq \gamma_1 \geq \lambda_2 \geq \gamma_2 \geq \ldots \geq \gamma_{N-1} \geq \lambda_N.$$

As a consequence, for any $z \in \mathbb{H}$,

$$\frac{1}{N-1} \operatorname{Tr} G^{[i]}(z) = \frac{1}{N} \operatorname{Tr} G(z) + o_p(1).$$

Proof. The first part is the Cauchy Interlace Theorem, a standard result in matrix analysis, see [Hwa04]. Let $f(x) = \frac{1}{x-z}$ for $x \in \mathbb{R}$. Then,

$$\frac{1}{N} \operatorname{Tr} G(z) - \frac{1}{N-1} \operatorname{Tr} G^{[i]}(z) = \frac{f(\lambda_N)}{N} - \frac{1}{N} \sum_{i=1}^{N-1} (f(\lambda_i) - f(\gamma_i)) + \left(\frac{1}{N} - \frac{1}{N-1}\right) \sum_{j=1}^{N-1} f(\gamma_j).$$

Since $f(x) \leq \frac{1}{\operatorname{Im} z}$, we have that

$$\left| \left(\frac{1}{N} - \frac{1}{N-1}\right) \sum_{j=1}^{N-1} f(\mu_j) \right| \leq \left| \frac{1}{N} - \frac{1}{N-1} \right| \cdot \frac{N-1}{\operatorname{Im} z} = \frac{1}{N \operatorname{Im} z} = o_N(1).$$

As for the second term, we first note that f is Lipschitz continuous with constant $L = (\operatorname{Im} z)^{-2}$ since for any $x, y \in \mathbb{R}$,

$$|f(x) - f(y)| = \left| \frac{1}{x-z} - \frac{1}{y-z} \right| = \frac{|x-y|}{|x-z||y-z|} \leq \frac{|x-y|}{(\operatorname{Im} z)^2}.$$

By the fact that $\gamma_i \in [\lambda_{i+1}, \lambda_i]$,

$$|f(\gamma_i) - f(\lambda_i)| \leq L(\lambda_i - \gamma_i) \leq L(\lambda_i - \lambda_{i+1}),$$

we have

$$\frac{1}{N} \sum_{i=1}^{N-1} f(\lambda_i) - f(\gamma_i) \leq \frac{L}{N} \sum_{i=1}^{N-1} (\lambda_i - \lambda_{i+1}) = \frac{L}{N}(\lambda_N - \lambda_1).$$

Since both $|\lambda_N| \leq ||H||_{\text{op}}$ and $|\lambda_1| \leq ||H||_{\text{op}}$, we have that $\lambda_N - \lambda_1 \leq 2||H||_{\text{op}}$, and it suffices to bound the operator norm. This probabilistic bound is shown in greater generality in [FK81], and gives that $\lambda_N - \lambda_1 \xrightarrow{p} 4$, hence this second term is $o_p(1)$.

Finally, the first term has $\frac{f(\lambda_N)}{N} \leq \frac{1}{N \operatorname{Im} z}$, and thus goes to 0. Since all terms involved have limit 0 in probability as $N \to \infty$, the proof is complete.

□

Updating the previous equation using Lemma 3.4,

$$G_{11} = -\frac{1}{z + \frac{1}{N}\operatorname{tr} G + o_p(1)}.$$

Since we assumed $i = 1$ without loss of generality, we obtain the normalized trace

$$\frac{1}{N}\operatorname{tr} G = -\frac{1}{z + \frac{1}{N}\operatorname{tr} G + o_p(1)}.$$

By Proposition 2.6, we have the asymptotic self-consistent equation

$$m_N(z) = -\frac{1}{z + m_N(z) + o_p(1)}.$$

For any fixed z, we have that $|m_N(z)| \leq (\operatorname{Im} z)^{-1}$ since $|\lambda_i - z| \geq \operatorname{Im} z$ for all eigenvalues. This bound is constant with respect to N, hence rearranging the above equation we obtain

$$1 + m_N(z)(z + m_N(z)) \xrightarrow{p} 0,$$

pointwise. It remains to show that this asymptotic self-consistent equation is stable, in the sense that any sequence (m_N) satisfying $1 + m_N(z)(z + m_N(z)) \xrightarrow{p} 0$ must converge to the unique solution m of the equation $1 + m(z)(z + m(z)) = 0$.

3.2 Stability Step

Let m satisfy the exact equation $1 + m(z)(z + m(z)) = 0$.

Lemma 3.5 (Exact Solution). *For all $z \in \mathbb{H}$, the unique solution to $1 + m(z)(z + m(z)) = 0$ is given by*

$$m(z) = \frac{-z + \sqrt{z^2 - 4}}{2},$$

whose corresponding inverse Stieltjes transform has density

$$\rho(z) = \frac{1}{2\pi}\sqrt{(4 - x^2)_+}, \quad x \in \mathbb{R}.$$

Proof. Solving the exact equation using the quadratic formula yields two solutions

$$m(z) = \frac{-z \pm \sqrt{z^2 - 4}}{2}.$$

First, consider the positive branch. By using the Taylor expansion of the square root,

$$m_+(z) = \frac{-z + z - 2/z + O(z^{-3})}{2} = -\frac{1}{z} + O(|z|^{-3}).$$

Letting $z = x + i\eta$ with $x \in \mathbb{R}$ and $\eta > 0$,

$$\operatorname{Im}\left(-\frac{1}{z}\right) = -\operatorname{Im}\left(\frac{x - i\eta}{x^2 + \eta^2}\right) = \frac{\eta}{x^2 + \eta^2} > 0.$$

Hence, we have $\operatorname{Im} m_+(z) > 0$ for large enough z. For the negative branch, a similar analysis yields

$$m_-(z) = -z + O(z^{-1}),$$

so $\operatorname{Im} m_-(z) < 0$ for large enough z. Since each branch is clearly analytic, if $\operatorname{Im} m(z)$ changed sign inside $\mathbb{H}$, we would necessarily have $\operatorname{Im} m(z) = 0$ for some $z \in \mathbb{H}$. Let $y \in \mathbb{R}$ be the value of $m(z)$ at this point. From the exact equation, we have

$$m(z) = -\frac{1}{z + m(z)},$$

which would imply that

$$\operatorname{Im} m(z) = -\operatorname{Im}\left(\frac{1}{x + i\eta + y}\right) = -\operatorname{Im}\left(\frac{x + y - i\eta}{(x + y)^2 + \eta^2}\right) = \frac{\eta}{(x + y)^2 + \eta^2} > 0,$$

which contradicts that $\operatorname{Im} m(z) = 0$. Thus, $m(z)$ does not change signs. Since we must have $\operatorname{Im} m(z) > 0$ for all $z \in \mathbb{H}$ by a property of the Stieltjes transform, the positive branch is therefore the unique solution.

Using Proposition 2.3,

$$\frac{1}{\pi}\lim_{\eta\to 0^+}\int_a^b \operatorname{Im}\left(\frac{-(x+i\eta)+\sqrt{(x+i\eta)^2-4}}{2}\right)dx = \frac{1}{2\pi}\int_a^b \sqrt{(4-x^2)_+}\,dx = \mu((a,b)).$$

Since a finite Borel measure on $\mathbb{R}$ is uniquely determined by its values on intervals, it follows that μ as precisely the probability measure with density $\rho(x) = \frac{1}{2\pi}\sqrt{(4-x^2)_+}$. □

Let $\varepsilon_N(z) = 1 + m_N(z)(z + m_N(z))$, where $\varepsilon_N(z) \xrightarrow{p} 0$ pointwise, as we have shown. Let $\delta_N(z) = m_N(z) - m(z)$. For simplicity, we drop the parameter z from relevant quantities as well as the mode of convergence when obvious by context. We wish to show that for any $z \in \mathbb{H}$, if $\varepsilon_N \to 0$ then $\delta_N \to 0$.

By subtracting the exact equation from this equation, we obtain

$$\varepsilon_N = m_N(z+m_N) - m(z+m) = z\delta_N + \delta_N(\delta_N + 2m) = (z+2m)\delta_N + \delta_N^2.$$

Furthermore, rearranging the exact equation gives $z = -\frac{m^2+1}{m}$, and hence

$$\varepsilon_N = \frac{m^2-1}{m}\delta_N + \delta_N^2.$$

In order to establish convergence, we first show a bound $|\delta_N| \le C\,|\varepsilon_N|$ for some $C > 0$ potentially depending on z for a deterministic sequence (m_N).

Lemma 3.6. *For any fixed $z \in \mathbb{H}$, there exists a $c(z) > 0$ such that*

$$\left|\frac{(m(z))^2 - 1}{m(z)}\right| \ge c(z) > 0.$$

Furthermore, if (δ_N) is a deterministic sequence, there exists an $N_0 \ge 1$ such that for all $N \ge N_0$,

$$|\delta_N(z)| \le \frac{c(z)}{2} \quad \textit{and} \quad |\delta_N(z)| \le \frac{2}{c(z)}\,|\varepsilon_N(z)|.$$

Proof. Let $s(z) = \sqrt{z^2-4}$. Rearranging the exact solution for m found in Lemma 3.5, we have $2m + z = s$ From the exact Dyson equation $1 + m(z+m) = 0$, we also have $m + \frac{1}{m} = -z$. Thus,

$$\frac{m^2-1}{m} = 2m - \left(m + \frac{1}{m}\right) = 2m + z = s(z).$$

Let $c(z) = \min\{|z-2|, |z+2|\}$. For $c(z) > 0$,

$$|s(z)|^2 = |z^2 - 4| = |z-2||z+2| \ge c(z)^2.$$

Thus, $|s(z)| \ge c(z) > 0$. The case $c = 0$ is more involved and is treated in greater generality in [EY17].

Assume for contradiction that for all N_0, there exists an $N_* > N_0$ such that $|\delta_N| > \frac{c}{2}$. By the reverse triangle inequality,

$$|\varepsilon_{N_*}| \geq \left|\frac{m^2-1}{m}\right| |\delta_{N_*}| - |\delta_{N_*}|^2 \geq |\delta_{N_*}|(c - |\delta_{N_*}|) > \frac{c^2}{4}.$$

By assumption, $\varepsilon_N \to 0$, which contradicts the previous claim. Now that we have established that there is some N_0 such that $|\delta_N| \leq c/2$ for $N \geq N_0$, a slight modification to the valid part of the previous inequality gives

$$|\delta_N| \leq \frac{|\varepsilon_N|}{c - |\delta_N|} \leq \frac{2}{c}|\varepsilon_N|,$$

for all $N \geq N_0$. □

Since $c(z) > 0$ is independent of N and $\varepsilon_N(z) \to 0$, it follows that $\delta_N(z) \to 0$, and thus $m_N \to m$ pointwise. While this was done for a deterministic sequence $m_N(z)$, the same analysis can be carried out in the probabilistic setting, up to minor modifications, yielding $m_N \xrightarrow{p} m$ pointwise. By Proposition 2.4 applied to the exact solution found in Lemma 3.5, we have that the sequence of empirical spectral distributions μ_N converges weakly to the measure μ with density ρ. This concludes the proof of Theorem 3.2.

4 Numerical Experiments

4.1 Empirical Spectral Distribution

We consider the following sequences of Wigner matrices:

1. (A_N) with entries $A_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1/N)$,
2. (B_N) with i.i.d. entries $B_{ij} = \pm 1/\sqrt{N}$, each with probability $1/2$,
3. (C_N) with entries $C_{ij} = (Y_{ij} - 1)/\sqrt{N}$, where $Y_{ij} \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$.

Observe that the distributions of the entries of A_N, B_N, and C_N, are continuous, discrete, and non-symmetric, respectively. We define the histogram density approximation of the empirical spectral measure, given by

$$\hat{\rho}(B_k) = \frac{\mu_N(B_k)}{|B_k|},$$

where $(B_k)_k$ is a partition of $\mathbb{R}$ into bins. Using appropriately chosen bin sizes, Figure 1 displays $\hat{\rho}$ for a single realization of A_N (blue), B_N (red), and C_N (green), for $N = 50$, 500, 5000. These results provide strong evidence for the semicircle law.

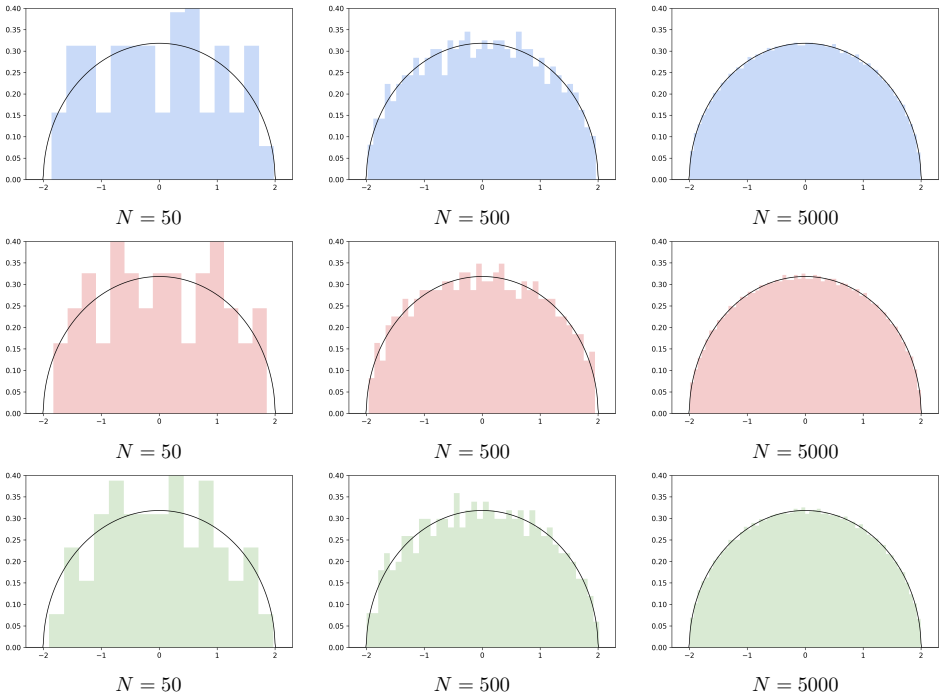


Figure 1: Convergence of $\hat{\rho}$ to the semicircle law.

4.2 Stieltjes Transform

Let $z = 1 + i\eta$, where $\eta = 0.1, 0.25, 0.5, 1.0, 2.0$. For evenly spaced values of N between 50 to 2000, we compute $|\hat{\delta}_N|$, the sample mean of $|\delta_N| = |m_N(z) - m(z)|$, calculated using 10 realizations of A_N. Figure 2 plots $\log(|\hat{\delta}_N|)$ as a function of N for each choice of η.

Although not shown directly, several calculations in the preceding analysis suggest that δ_N should scale $O\left(\frac{1}{N\eta}\right)$. As a consequence, we should expect that $\log \delta_N \approx -\log N - \log \eta + C$, whose behaviour is consistent with the results in Figure 2. We expect similar behaviour for other choices of $\operatorname{Re} z$ as η varies.

4.3 Dyson Equation

Lemma 3.6 establishes a bound of the form $|\delta_N| \leq C|\varepsilon_N|$ for sufficiently large N, where $C = 2/\operatorname{dist}(z, \{-2, 2\})$. In Figure 3, we illustrate the log-log relationship between $|\hat{\varepsilon}_N|$ and $|\hat{\delta}_N|$ for three values of z alongside the corresponding theoretical bounds, over a range of N values from 50 to 1000 with equal spacing.

The seemingly linear relationship is likely due to the fact that we have the exact equality

$$\varepsilon_N = \frac{m^2 - 1}{m}\delta_N + \delta_N^2,$$

which is roughly linear for $\delta_N \approx 0$.

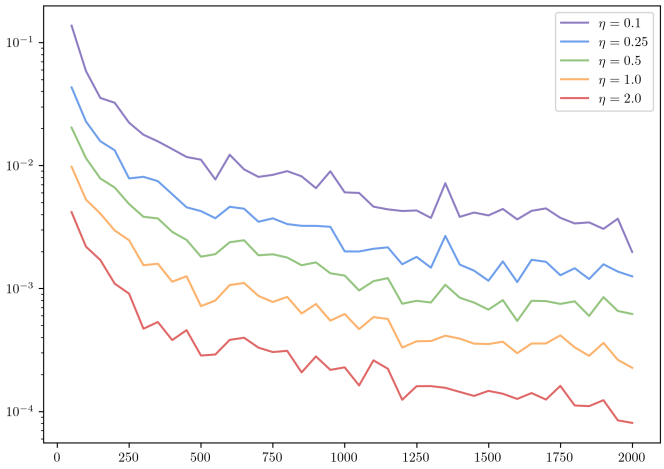


Figure 2: Convergence of $|\hat{\delta}_N|$.

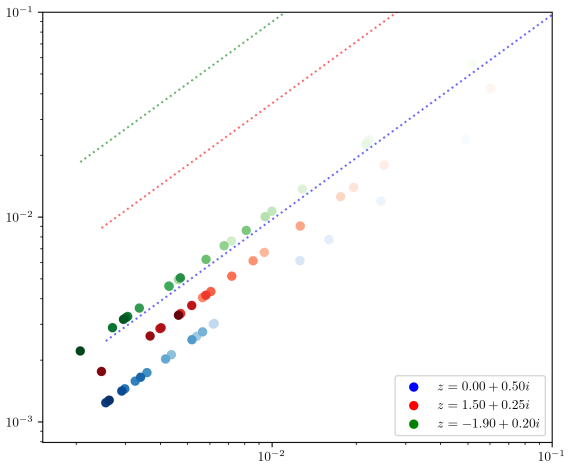


Figure 3: Convergence and boundedness of $|\hat{\delta}_N|$ in relation to $|\hat{\varepsilon}_N|$ on a log-log scale. Dotted lines represent bounds. Opacity of points increases as N increases.

5 Conclusion

While we presented the Dyson equation for the classical Wigner ensemble, the most general framework extends well beyond this case. Our analysis is primarily based on the initial parts of [Erd19], and the remainder of that work develops this more general approach to deriving and using Dyson equations, notably for the case of correlated entries or potentially non-uniform variance profiles. The Dyson equation can also be used to derive the Marchenko–Pastur law for eigenvalues of sample covariance matrices, as shown in [Paq24].

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